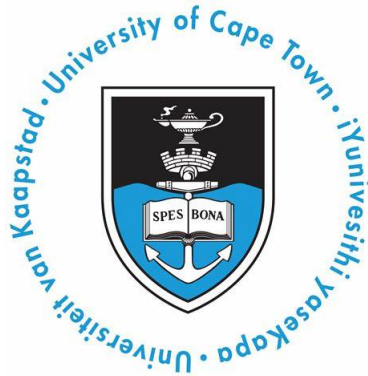




UNIVERSITY OF CAPE TOWN

COURSE CODE

STA5076Z

Predicting Medical Costs using Multiple Linear Regression

Author:
Lungile P. Nkuna

Student Number:
NKNPRA001

25 April 2025

Contents

1	Introduction	2
2	Data Wrangling	2
3	Exploratory Data Analysis (EDA)	3
4	Model Building	6
4.1	Multiple linear regression	6
4.1.1	Model fit performance	6
4.1.2	Significance of Individual Predictor Variables	7
4.2	Prediction and MSE calculation	7
5	Model Improvement	8
5.1	Variable Selection Techniques	8
5.1.1	Backwards stepwise	8
5.1.2	Ridge and Lasso	10
5.2	Prediction and MSE Calculation	11
5.3	Interaction Term	12
5.4	Prediction and MSE Calculation	12
6	Residual Diagnostics	13
7	Conclusion	15
7.1	Practical Implications: Factors Contributing to Medical Costs	15
7.2	Impact of the Interaction Term Between Smoking and BMI	15
8	Appendix	16

1 Introduction

This study aims to investigate the factors influencing individual medical charges billed by health insurance providers in the United States. The analysis is based on a dataset comprising 500 observations and 7 variables. The primary objective is to develop a linear regression model to predict individual medical costs (charges) using six predictor variables: sex, age, BMI, number of children, smoking status, and region.

To achieve this objective, several analytical steps were undertaken. The process began with exploratory data analysis (EDA) to examine the distribution of variables and explore potential relationships among them. Following EDA, a full regression model was developed, incorporating all six predictors. The model was further refined using variable selection techniques such as stepwise regression, as well as regularisation methods such as Ridge and Lasso regression, to improve model accuracy and interpretability.

Model performance was evaluated using the mean squared error (MSE), ensuring robust assessment of predictive capability. Additionally, interaction terms were introduced in the final model to capture more complex relationships between variables. Finally, residual diagnostics were conducted to assess the validity of the underlying assumptions of linear regression.

The findings from this analysis provide valuable insights into the key factors driving medical insurance charges in the US. The conclusions also highlight the practical implications of these findings and offer meaningful interpretations of the model's outcomes.

2 Data Wrangling

The dataset was carefully inspected for missing values and to verify the data types of each variable before performing exploratory data analysis (EDA). It was confirmed that there were no missing values in the dataset, ensuring that no imputation or removal of records was necessary prior to analysis.

The variables were found to have the following data types: BMI and charges were numeric, while sex, smoker, and region were stored as character variables. The variables children and age were recorded as integers. In preparation for building the linear model (lm), the categorical variables (sex, smoker, region, and children) were converted to factors. This step ensures that the linear model correctly treats these variables as categorical predictors rather than continuous numeric inputs, without negatively impacting the model's performance. By properly formatting the dataset before analysis, the modelling process was able to accurately capture the influence of categorical variables and produce more reliable results.

3 Exploratory Data Analysis (EDA)

	Charges	Children	BMI	Age
Min.	1132	0.00	15.96	18.00
1st Qu.	4565	0.00	26.41	25.00
Median	9274	1.00	30.21	38.00
Mean	12769	1.01	30.66	38.23
3rd Qu.	16249	2.00	34.80	50.00
Max.	58571	5.00	53.13	64.00

Table 1: Summary statistics for numerical values

The distribution of charges is right-skewed, with most individuals paying moderate amounts and a few paying extremely high costs. The minimum charge is \$1,132, the maximum is \$58,571, the median is \$9,274, and the mean is higher at \$12,769, reflecting the impact of high outliers. Most individuals have between 0 and 2 children, with a median of 1 and a maximum of 5 children. BMI values range widely from 15.96 to 53.13, indicating that the sample includes both underweight and severely overweight individuals. The ages of the participants range from 18 to 64 years, with a median age of 38, suggesting that the sample covers both young adults and middle-aged individuals.

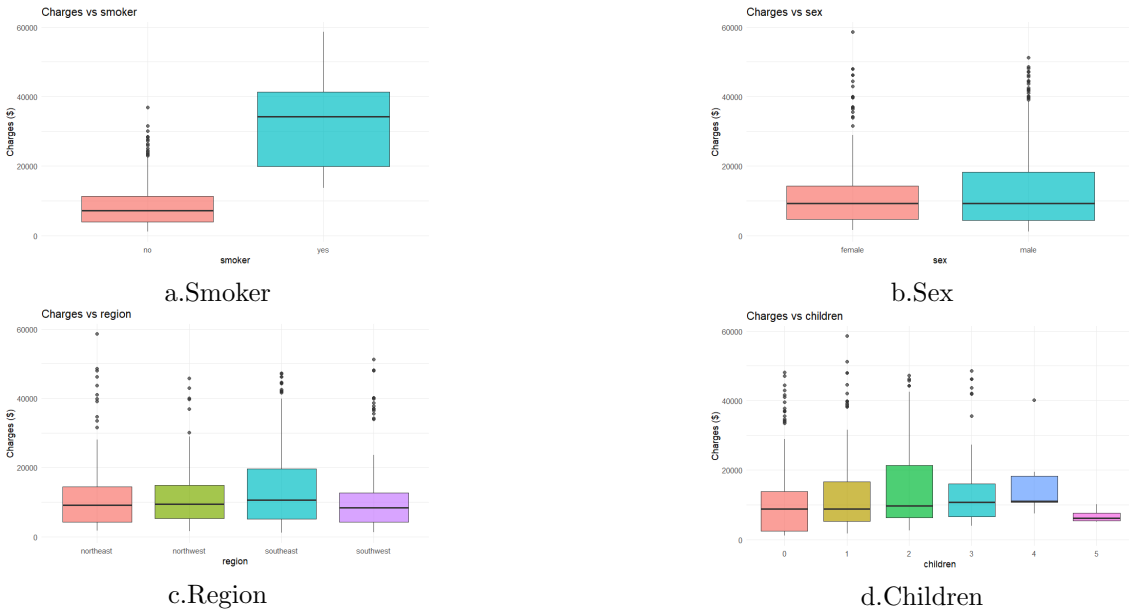


Figure 1: Distribution of data between charges and predictor variables

Smoking status has a strong influence on insurance charges (Figure 1a). Smokers have much higher charges, with most smoker charges falling between \$20,000 and \$45,000, compared to non-smokers whose charges mostly stay below \$15,000. Outliers above \$40,000 are common among smokers. Although there is a slightly wider spread for males, sex (Figure 1b) in general does not have a strong impact as both males and females generally have charges under \$20,000, although some individuals in both groups appear as outliers above \$40,000. Charges are relatively similar across regions (Figure 1c), but median charges are slightly higher in the Southeast and Northeast. All regions, however, show similar spreads and multiple high outliers above \$30,000. The number of

children (Figure 1d) does not strongly influence charges and shows no clear trend; whether an individual has 0, 1, or more children, the typical charges remain similar. Outliers are present across all child groups, particularly for individuals with 0–3 children. Overall, the data reveal a strong pattern between smoking and medical costs, a right-skewed distribution with many extreme values, and relatively weak effects for sex, region, and number of children.

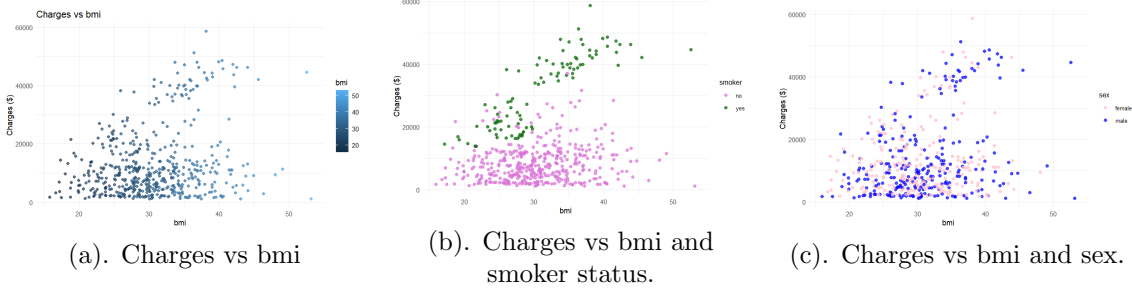


Figure 2: Distribution of charges and BMI filtered by sex and smoker status

Figure 2a shows that there is a positive but weak association between BMI and insurance charges. As the BMI increases, the charges show a slight tendency to rise; however, the relationship is highly scattered. In particular, there are outliers at higher BMI and charge values, indicating that while not all people with a high BMI incur large charges, when they do, the amounts can be extremely substantial, reaching up to \$60,000. In Figure 2b, separating individuals by smoker status reveals a much stronger pattern. Smokers (green dots) have significantly higher charges than non-smokers (purple dots) for the same BMI levels. Most high-charge outliers are smokers, with some smokers who have a BMI greater than 30 to 35 incurring charges close to \$60,000. In contrast, Figure 2c, which filters by sex, shows that the difference between males and females is much less pronounced. Both sexes exhibit a similar spread of charges across BMI values, suggesting that sex does not have a strong impact on insurance charges when compared to BMI or smoking status.

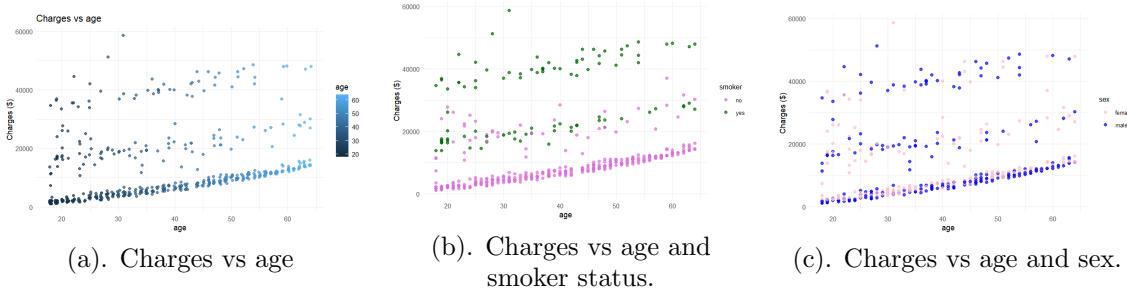


Figure 3: Distribution of charges and age filtered by sex and smoker status

Figure 3a shows a strong positive relationship between age and insurance charges. As age increases from 18 to 64 years, insurance charges generally increase. Younger individuals (around 18–30 years) mostly have charges below \$15,000, while older individuals (above 50 years) more often have charges above \$15,000. There are some noticeable outliers, with charges reaching as high as \$60,000 across different age groups. In Figure 3b, separating by smoking status reveals an even stronger positive trend. Smokers (green dots) consistently have higher charges than non-smokers (purple dots) at all ages. Among smokers aged 50 and older, many charges range between \$25,000 and \$50,000, while most non-smokers typically remain below \$20,000 even at older ages. Most extreme

outliers, with charges exceeding \$40,000, are smokers. In Figure 3c, when comparing males and females, both groups show a similar increasing trend with age. Males have a slightly wider spread of charges after age 40, but both sexes mainly have charges below \$30,000 before 50 years old, and higher spreads after 50. Overall, the data show that age and smoking status have a strong, quantifiable effect on medical charges, while sex plays a relatively minor role.

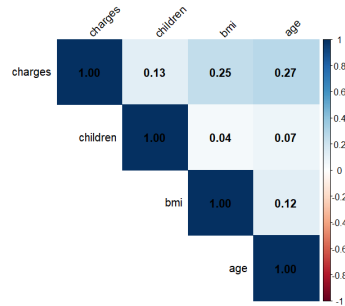


Figure 4: Correlation matrix for numerical values

Figure 4 presents the correlation matrix for the numerical variables: charges, children, BMI, and age. The results show that charges have the strongest positive correlation with age (0.27), meaning that as individuals get older, their insurance charges tend to increase moderately. There is also a weaker but still positive correlation between charges and BMI (0.25), suggesting that individuals with higher BMI generally incur slightly higher insurance charges. The correlation between charges and number of children is very weak (0.13), indicating that the number of dependents has minimal direct impact on insurance costs. Looking at the other variables, BMI and age have a very weak positive correlation (0.12), while children show almost no meaningful correlation with BMI (0.04) or age (0.07). Overall, age is the most influential numerical factor associated with charges, followed by BMI, whereas number of children has little to no effect.

4 Model Building

4.1 Multiple linear regression

In this analysis, a multiple linear regression model was developed to predict medical insurance charges based on various features such as age, sex, body mass index (BMI), number of children, smoking status, and region. The dataset was first split into training and testing sets, with 80% of the data allocated for training the model and 20% reserved for testing its performance. The regression model was fitted using the training data (train.data), with charges as the response variable and all other available variables as predictors. The model output includes the estimated coefficients for each predictor, along with their standard errors, t-values, and p-values, which assess the significance of each feature in predicting insurance charges.

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-10409.55	1825.81	-5.701	2.36e-08 ***
age	235.24	22.08	10.655	$< 2 \times 10^{-16}$ ***
sexmale	-274.52	618.02	-0.444	0.6571
bmi	308.36	53.79	5.733	1.99e-08 ***
children1	1240.58	765.38	1.621	0.1059
children2	2179.72	871.81	2.500	0.0128 *
children3	1995.04	1065.82	1.872	0.0620 .
children4	4091.59	2363.16	1.731	0.0842 .
children5	2682.26	3117.34	0.860	0.3901
smokeryes	23368.14	789.43	29.601	$< 2 \times 10^{-16}$ ***
regionnorthwest	-386.68	894.25	-0.432	0.6657
regionsoutheast	-1010.18	866.71	-1.166	0.2445
regionsouthwest	-862.83	847.47	-1.018	0.3092

Significance codes: 0 '***', 0.001 '**', 0.01 '*', 0.05 '.', 0.1 ' ', 1.

Residual standard error: 6067 on 387 degrees of freedom.

Multiple R-squared: 0.7342, Adjusted R-squared: 0.726.

F-statistic: 89.08 on 12 and 387 DF, p-value: $< 2.2 \times 10^{-16}$.

lm(formula = charges ~ ., data = train.data)

Table 2: Linear regression results predicting **charges**.

4.1.1 Model fit performance

The multiple linear regression model demonstrates a good fit, as indicated by the multiple R-squared value of 0.7342, which suggests that approximately 73.42% of the variance in medical insurance charges is explained by the predictors included in the model. This is a strong result, indicating that the model is able to capture a significant portion of the underlying variability in the target variable. The adjusted R-squared value of 0.726, though slightly lower, accounts for the number of predictors in the model, reflecting a marginal decrease in explanatory power when accounting for model complexity.

The F-statistic of 89.08, with an associated p-value less than 2.2×10^{-16} , confirms that the model is statistically significant overall. This suggests that at least one of the predictors is contributing meaningfully to the prediction of insurance charges. Additionally, the residual standard error of 6067 indicates that, on average, the predictions deviate from the actual values by 6067 units.

4.1.2 Significance of Individual Predictor Variables

The significance of individual predictors is assessed through their respective t-values and p-values. A p-value below 0.05 typically indicates statistical significance. First, the intercept term is highly significant with a p-value of 2.36×10^{-8} . Among the predictors, age stands out with a highly significant positive relationship with insurance charges.

The coefficient for age is 235.24, with a p-value of less than 2×10^{-16} , indicating that older individuals tend to have higher insurance charges. This is a common expectation in the insurance industry where age determines premiums.

The variable `sexmale`, however, does not appear to significantly influence insurance charges, as evidenced by a coefficient of -274.52 and a p-value of 0.6571. This suggests that, in this dataset, sex does not contribute meaningfully to the variation in insurance charges. In contrast, BMI is highly significant, with a coefficient of 308.36 and a p-value of 1.99×10^{-8} . This suggests that higher BMI is associated with significantly higher charges, which reflects the well-established link between body weight and health-related risks, influencing insurance premiums.

The number of children a policyholder has also affects insurance charges, though not as strongly as other predictors. The coefficient for `children2` (2 children) is 2179.72, with a p-value of 0.0128, indicating a statistically significant positive effect on charges. However, the impact of other child categories, such as `children3` and `children4`, is more marginal, with p-values slightly above the 0.05 threshold. This suggests that the presence of children influences charges, though the effect diminishes with higher numbers of children.

The `smokeryes` variable exhibits a particularly strong relationship with insurance charges, with a coefficient of 23,368.14 and a p-value of less than 2×10^{-16} . This indicates that smokers face significantly higher premiums than non-smokers, which is consistent with actuarial models that account for the higher health risks associated with smoking. Finally, the regional variables (`regionnorthwest`, `regionsoutheast`, `regionsouthwest`) do not show significant effects on insurance charges, as all have p-values greater than 0.05. This suggests that, within the context of this model, the region in which a person resides does not significantly impact their insurance premium.

4.2 Prediction and MSE calculation

After creating the linear regression model, it was used to predict medical costs for new, unseen data. This involved applying the model to a separate dataset (`test.data`), which resulted in the generation of predicted insurance charges for the observations within this dataset. Following model testing, the next step was to compute the Mean Squared Error (MSE) of the linear regression model. The MSE measures the average squared difference between the actual charges in the test dataset (`test.data$charges`) and the predicted charges.

The computed MSE value of 35,858,116 indicates that, on average, the squared difference between the actual and predicted charges in the testing set is 35,858,116. A higher MSE value suggests that the model's predictions deviate substantially from the actual values. In this case, this value reflects the overall accuracy and performance of the model on unseen data.

To interpret the MSE further, it provides a sense of the magnitude of prediction errors. A lower MSE would typically indicate better predictive performance, meaning the model's predictions are closer to the actual values. Conversely, a higher MSE suggests that the model's predictions are more off-target, implying that there is room for improvement in the model's accuracy.

5 Model Improvement

5.1 Variable Selection Techniques

5.1.1 Backwards stepwise

Stepwise selection is a technique used to identify the best subset of predictors in a model, minimising prediction error by iteratively adding or removing predictors. There are three main methods: forward selection, backward selection, and bidirectional stepwise selection.

In forward selection, the model starts with no predictors, and the most significant predictors are added iteratively. The process stops when further addition no longer improves the model significantly.

Backward selection starts with all predictors in the model and removes the least significant ones. It stops when all remaining predictors are statistically significant, and removing any more would reduce the model's performance.

Bidirectional stepwise selection combines both forward and backward methods, adding or removing predictors at each step, allowing for more flexibility in model refinement..

To assess the models, we use Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). AIC balances model fit with complexity, aiming to minimise the value. BIC works similarly but applies a stronger penalty for complexity. Both are used to avoid overfitting while selecting the best model. For this assignment, we will use backward selection, as we have already fitted a full model. AIC and BIC will guide us in selecting the optimal predictors, with the model that has the lowest values of these criteria being preferred.

AIC

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-10576.22	1712.07	-6.177	1.64e-09 ***
age	235.66	21.92	10.752	< 2 × 10 ⁻¹⁶ ***
bmi	291.28	50.73	5.742	1.88e-08 ***
children1	1173.05	758.33	1.547	0.1227
children2	2160.77	865.04	2.498	0.0129 *
children3	1936.73	1060.74	1.826	0.0686 .
children4	4214.45	2346.16	1.796	0.0732 .
children5	2351.05	3070.45	0.766	0.4443
smokeryes	23295.21	780.63	29.842	< 2 × 10 ⁻¹⁶ ***

Significance codes: 0 '***', 0.001 '**', 0.01 '*', 0.05 '.', 0.1 ' ', 1.

Residual standard error: 6051 on 391 degrees of freedom.

Multiple R-squared: 0.7329, Adjusted R-squared: 0.7274.

F-statistic: 134.1 on 8 and 391 DF, p-value: < 2.2 × 10⁻¹⁶.

lm(formula = charges ~ age + bmi + children + smoker, data = train.data)

Table 3: Backward stepwise regression results using AIC.

The intercept of -10,576.22 represents the expected charge when all predictors (age, BMI, number of children, and smoker status) are zero. While this value is not practically meaningful, it serves as the baseline for the model.

The age coefficient of 235.66 means that for each additional year of age, the charge increases by 235.66 units, with a highly significant p-value ($< 2e-16$), indicating a strong relationship between age and charges.

The BMI coefficient of 291.28 suggests that each unit increase in BMI leads to a charge increase of 291.28 units. This effect is also statistically significant ($p < 2e-16$), emphasising the importance of BMI in predicting charges.

For the number of children, the effect varies: Having 2 children increases the charge by 2,160.77 units, which is significant ($p = 0.0129$). Having 3, 4, or 5 children shows weaker or inconsistent effects ($p > 0.05$), suggesting diminishing returns beyond 2 children.

The smoker status coefficient of 23,295.21 indicates that smokers face significantly higher charges than non-smokers, with a very low p-value ($< 2e-16$), underlining the major impact of smoking.

The R-squared value of 0.7329 means the model explains 73.3% of the variance in charges, indicating a good fit. The adjusted R-squared of 0.7274 confirms that the model remains robust after adjusting for the number of predictors.

The F-statistic of 134.1 with a p-value $< 2.2e-16$ shows that the model is statistically significant and the predictors together effectively explain the variability in charges.

BIC

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-10212.32	1700.90	-6.004	4.36e-09 ***
age	238.97	21.97	10.880	$< 2 \times 10^{-16}$ ***
bmi	305.48	50.47	6.053	3.30e-09 ***
smokeryes	23418.78	782.67	29.922	$< 2 \times 10^{-16}$ ***

Significance codes: 0 '***', 0.001 '**', 0.01 '*', 0.05 '.', 0.1 ' ', 1.

Residual standard error: 6093 on 396 degrees of freedom.

Multiple R-squared: 0.7257, Adjusted R-squared: 0.7236.

F-statistic: 349.2 on 3 and 396 DF, p-value: $< 2.2 \times 10^{-16}$.

lm(formula = charges ~ age + bmi + smoker, data = train.data)

Table 4: Backward stepwise regression results using BIC.

The intercept of -10,212.32 represents the baseline charge when all predictors (age, BMI, and smoker status) are zero. While this value is mathematically valid, it does not hold practical significance as these predictors cannot realistically be zero.

The age coefficient of 238.97 indicates that, for each additional year of age, the expected charge increases by 238.97 units. This relationship is highly significant $p < 2 \times 10^{-16}$, suggesting that age strongly influences the charges.

The BMI coefficient of 305.48 shows that each unit increase in BMI leads to a charge increase of 305.48 units. This effect is statistically significant ($p = 3.30e-09$), highlighting the importance of BMI in determining charges.

The smoker status coefficient of 23,418.78 indicates that smokers face much higher charges than non-smokers. This relationship is highly significant ($p < 2e-16$), underlining the major role smoking plays in predicting charges.

The R-squared value of 0.7257 means that the model explains about 72.6% of the variance in charges, suggesting a good fit. The Adjusted R-squared of 0.7236 supports this, indicating the model remains strong after adjusting for the number of predictors.

The F-statistic of 349.2 with a p -value $< 2.2\text{e-}16$ shows that the overall model is highly statistically significant, meaning the predictors together provide a strong explanation for the variability in charges.

5.1.2 Ridge and Lasso

Ridge and lasso and penalized regression methods. The aim of these penalized regression methods is to reduce the coefficient values towards zero. This allows the less contributive variables to have a coefficient close to zero or equal to zero. The methods require a tuning parameter (lambda) that will determine the amount of shrinkage. Ridge regression shrinks the regression coefficients so that variables with minor contributions to the outcome have their coefficients close to zero with the penalty L2-norm, which is the sum of squared coefficients. The lasso shrinks the regression coefficients towards zero by penalising the regression model with the penalty term L1-norm, which is the sum of the absolute coefficients.

Lambda controls the shrinkage applied to the coefficients when fitting the model. In the ridge model, the optimal lambda was found to be 900.663, and for lasso, the optimal lambda was 198.608.

The Ridge regression model (Appendix: Table 7) with an optimal lambda of 900.663 shows the relationship between various predictors and charges. The intercept of -8,860.45 represents the baseline charge when all predictors are zero. Age, with a coefficient of 216.36, suggests that each additional year of age increases the charge by 216.36 units. Sex (male) has a small negative coefficient of -109.97, indicating that being male slightly reduces the expected charge, though this effect is relatively minor. BMI has a coefficient of 287.76, meaning each unit increase in BMI raises the charge by 287.76 units. The number of children also influences charges, with higher charges for families with more children, especially for 2, 3, and 4 children. Smokers face significantly higher charges, with a coefficient of 21,610.64. Regional variables (northwest, southeast, southwest) have small negative coefficients, indicating slight reductions in charges for individuals in those regions compared to others. This model suggests that while several factors contribute to charges, the impact of smoking and the number of children are particularly noteworthy.

The LASSO model (Appendix: Table 8) with an optimal lambda of 198.608 identifies key predictors that influence charges. The intercept of -8,804.57 represents the baseline charge when all other variables are zero. Age, with a coefficient of 223.93, suggests that each additional year of age increases the charge by 223.93 units. BMI, with a coefficient of 268.71, indicates that each unit increase in BMI raises the charge by 268.71 units. The number of children also significantly affects charges, with increasing charges for higher numbers of children, especially for 2, 3, and 4 children. Smokers face substantially higher charges, with a coefficient of 22,815.34. The model excludes sex and region variables, indicating these factors do not contribute significantly to predicting charges in this case.

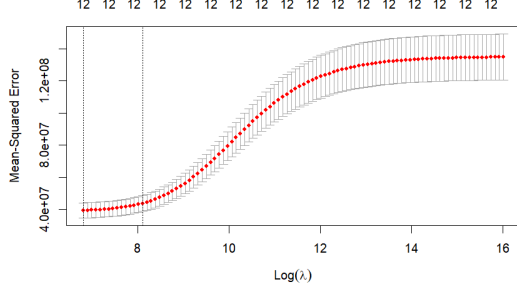


Figure 5: Ridge cross validation plot.

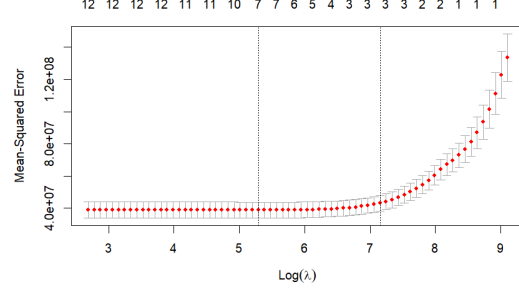


Figure 6: Lasso cross validation plot.

The optimisation process has identified distinct regularisation parameters for each model, with the Ridge regression yielding an optimal lambda of 900.663 and the LASSO model converging at lambda 198.608. These values emerge from the cross-validation plots, which visually demonstrate the relationship between prediction error and regularisation strength. The Ridge (Figure 5) model's higher lambda indicates substantial shrinkage of all coefficients toward zero without eliminating any variables completely, suggesting the model addresses multicollinearity while retaining all predictors. Meanwhile, the LASSO's (Figure 6) more moderate lambda value reflects its dual functionality of regularisation and feature selection, likely producing a sparser model by setting some coefficients exactly to zero. The cross-validation plots clearly illustrate these differences, with the Ridge plot showing a U-shaped error curve that minimises around $\log(\lambda)=10$ (corresponding to lambda 900.663) before gradually increasing, while the LASSO plot maintains relatively stable error rates until rising sharply after $\log(\lambda)=7$ (corresponding to lambda 198.608), confirming the selected optimal parameters. These lambda values represent the ideal balance between model complexity and predictive accuracy for each regression technique.

5.2 Prediction and MSE Calculation

After fitting the stepwise, Ridge, and Lasso regression models on the training data, the next step involved evaluating their performance on the testing dataset. To do this, the testing data was first converted into a matrix format, excluding the response variable, as required for penalized regression models. Each model was then applied to the testing data using the `predict()` function. Following prediction, the mean squared error (MSE) was calculated for each model to assess their predictive accuracy. These results provide insights into the models' effectiveness in predicting charges on previously unseen data.

Model	MSE
Full	35,858,116
Stepwise (AIC)	35,979,444
Stepwise (BIC)	37,126,266
Ridge	36,460,140
LASSO	36,909,829

Table 5: Comparison of Models based on Mean Squared Error (MSE)

Based on the comparison of mean squared error (MSE) values, the full model demonstrates the best predictive performance, achieving the lowest MSE of 35,858,116. This suggests that incorporating

all available predictors captures the underlying data patterns most effectively, resulting in minimal average prediction error. The stepwise (AIC) model follows closely, with an MSE of 35,979,444, indicating that it simplifies the model slightly while maintaining almost comparable predictive accuracy. In contrast, the stepwise (BIC) model yields a higher MSE of 37,126,266, reflecting BIC's tendency to favour more parsimonious models at the expense of slightly increased error. Ridge regression, with an MSE of 36,460,140, and LASSO regression, with an MSE of 36,909,829, both introduce regularisation techniques that are beneficial for handling multicollinearity and improving model interpretability; however, they come with a moderate trade-off in predictive accuracy in this case. Therefore, purely based on minimising MSE, the full model is the preferred choice.

5.3 Interaction Term

During model development, interaction terms based on insights from exploratory data analysis (EDA) were selected. The EDA indicated that age and BMI had the strongest correlations with insurance charges (correlation coefficients of approximately 0.27 and 0.25, respectively). Visualisations also highlighted that smoking status had the most substantial individual effect on charges, with most smokers incurring costs between \$20,000 and \$58,000, while most non-smokers typically paid less than \$15,000. Given these findings, interactions involving smoking were prioritised. Initially, the interaction between smoking and age was considered, as both were independently important; however, I also investigated the interaction between smoking and BMI due to the observation that smokers with higher BMI values often experienced exceptionally high charges.

The regression results reinforced the decision to focus on smoking-related interactions. In the model, smoking exhibited the largest individual effect on charges, with smokers predicted to pay approximately \$23,368 more than non-smokers, controlling for other variables. This was consistent with the EDA patterns. Other predictors, such as age and BMI, also showed statistically significant positive effects (\$235 and \$308 per unit increase, respectively), although their impacts were notably smaller than smoking. By contrast, variables such as sex and number of children had weaker and less consistent relationships with charges. Overall, the EDA and model findings together justified prioritising smoker–BMI as a key interaction term to capture the combined influence of health risks on insurance costs.

5.4 Prediction and MSE Calculation

After identifying the best model in this case, the full regression model predictions were generated for the test dataset using this model. The `predict()` function was utilised to estimate medical costs for each observation in the test set based on the fitted model. These predictions represent the model's estimated medical expenses for individuals not seen during training. Following the prediction step, the Mean Squared Error (MSE) was calculated to assess model performance, specifically for the version of the model that included interaction terms (smoker and BMI). The inclusion of an interaction term captures the combined effect of two predictor variables on the response variable, allowing for a more nuanced understanding of how predictors jointly influence insurance charges.

The computed MSE for the model with the smoker–BMI interaction was 26,490,736, which was an improvement compared to the model without interaction terms. This suggests that including the smoker–BMI interaction enhanced the model's ability to predict medical charges, highlighting the importance of accounting for joint effects between risk factors in modelling medical costs.

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1520.98	1604.49	-0.948	0.3437
age	235.16	17.88	13.151	$< 2 \times 10^{-16}$ ***
sexmale	-878.44	502.35	-1.749	0.0811 .
bmi	38.39	47.49	0.808	0.4194
children1	1193.39	619.93	1.925	0.0550 .
children2	1590.88	707.33	2.249	0.0251 *
children3	1077.73	865.65	1.245	0.2139
children4	4650.44	1914.45	2.429	0.0156 *
children5	1627.50	2525.98	0.644	0.5198
smokeryes	-24139.97	3387.81	-7.126	$5.12e-12$ ***
regionnorthwest	-623.05	724.49	-0.860	0.3903
regionsoutheast	-1020.93	701.99	-1.454	0.1467
regionsouthwest	-1275.15	687.02	-1.856	0.0642 .
bmi:smokeryes	1537.56	107.67	14.280	$< 2 \times 10^{-16}$ ***

Significance codes: 0 '***', 0.001 '**', 0.01 '*', 0.05 '.', 0.1 ' ', 1.

Residual standard error: 4914 on 386 degrees of freedom.

Multiple R-squared: 0.8261, Adjusted R-squared: 0.8202.

F-statistic: 141 on 13 and 386 DF, p-value: $< 2.2 \times 10^{-16}$.

lm(formula = charges ~ . + smoker * bmi, data = train.data)

Table 6: Linear regression results with interacting terms.

6 Residual Diagnostics

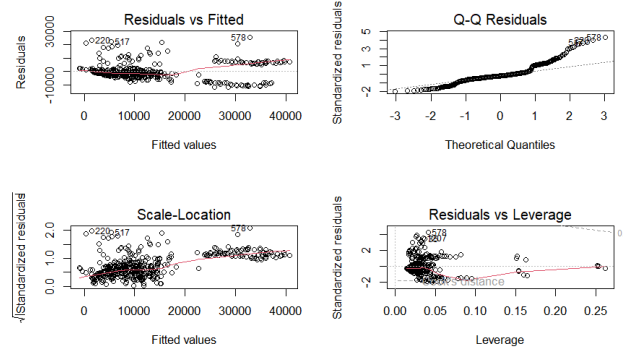


Figure 7: Correlation matrix for numerical values

Upon a thorough examination of the four diagnostic plots, it is evident that the current regression model significantly violates several key assumptions essential for valid statistical inference.

The Residuals vs Fitted plot reveals a clear departure from the expectation of random scatter around the zero line, with distinct clusters forming across various ranges of fitted values. This pattern suggests a failure to capture the true relationship between the predictors and the response variable, highlighting the potential presence of nonlinearities in the data that are not addressed by the current model specification. Consequently, this violates the assumption of linearity, indicating

that the model may be misspecified and overlooking important nonlinear components.

Further examination of the Residuals vs Fitted and Scale-Location plots provides compelling evidence of heteroscedasticity, which is a violation of the homoscedasticity assumption. Specifically, the variance of the residuals fluctuates substantially across the fitted values, which suggests that the error terms do not exhibit constant variance. This non-constant variance compromises the efficiency of the model's estimators and renders traditional hypothesis tests and confidence intervals invalid. Such a violation undermines the reliability of the statistical inferences derived from the model, necessitating remedial actions.

The Q-Q plot reveals notable deviations from the expected diagonal line, particularly in the tails, indicating that the residuals exhibit non-normality. This departure from normality, characterised by heavier tails than would be anticipated under a normal distribution, signals that standard inferential procedures relying on the normality assumption may lead to misleading conclusions. The failure to meet the normality assumption may particularly affect hypothesis tests and confidence intervals, undermining the robustness of the regression analysis.

Finally, the Residuals vs Leverage plot highlights potential influential observations, specifically points 578, 517, and 207, which could disproportionately affect the model's fit. These influential points, when examined in conjunction with the other diagnostic plots, suggest that the data structure may be more complex than the model can accommodate. The clustering patterns observed across all diagnostic plots further imply that the model fails to account for the underlying data dynamics.

In light of these substantial violations, it is recommended that the model be reconsidered through potential remedial measures. These could include the transformation of variables, the inclusion of polynomial or interaction terms to capture non-linear relationships, or the exploration of alternative modelling techniques such as generalised linear models (GLMs) or non-parametric approaches. By addressing the issues identified through these diagnostic plots, the model's validity and the accuracy of its statistical inferences can be significantly improved, ensuring more reliable and robust conclusions.

7 Conclusion

7.1 Practical Implications: Factors Contributing to Medical Costs

The analysis reveals that smoking status, age, and BMI (in interaction with smoking) are the most influential factors affecting medical charges. Smoking has the largest standalone impact: individuals who smoke face significantly higher medical costs compared to non-smokers. In the full linear model, being a smoker is associated with an increase of approximately \$23,368 in medical charges, highlighting the severe financial burden smoking imposes on healthcare expenses.

Age is another strong predictor of higher costs. Across both models, each additional year of age corresponds to an approximate \$235 increase in medical charges, reflecting the growing healthcare needs as individuals get older. Meanwhile, BMI alone has a moderate impact in the full model, with each additional BMI point associated with an increase of about \$308. However, in the interaction model, the influence of BMI becomes highly dependent on smoking status (discussed below).

The number of children a person has also shows some impact: individuals with two or four children tend to have somewhat higher charges, though the effect is relatively modest compared to smoking and age. In contrast, sex (male vs. female) and region (geographic location within the dataset) do not significantly affect medical costs. These findings suggest that lifestyle factors (like smoking and weight) and demographic factors (like age) are far more critical drivers of medical expenses than gender or regional differences.

7.2 Impact of the Interaction Term Between Smoking and BMI

Including the interaction term between smoking and BMI dramatically changes the interpretation of how BMI affects medical costs. In the interaction model, the main effect of smoking (smokeryes) appears strongly negative, but this is because its effect must be understood in combination with BMI. Specifically, the coefficient for the interaction term (bmi:smokeryes) is highly positive: \$1,537.56 per unit increase in BMI.

This means that for non-smokers, BMI has a negligible effect on medical costs — each point of BMI increases charges by only about \$38, a relatively small amount. However, for smokers, the impact of BMI is dramatically magnified. Each additional point of BMI for a smoker leads to a \$1,576 increase in medical charges (adding the base BMI effect and the interaction effect).

In simple terms, BMI is not a serious cost driver unless a person smokes. Among smokers, being overweight or obese severely worsens medical costs, likely due to compounding health risks such as cardiovascular disease, respiratory issues, and metabolic disorders.

The inclusion of the interaction term significantly improves the model's performance. The adjusted R-squared rises from 0.726 (full model) to 0.820 (interaction model), indicating that the interaction model explains much more of the variation in charges. Additionally, the residual standard error decreases, confirming that the predictions made by the interaction model are more accurate.

8 Appendix

Variable	Coefficient
(Intercept)	-8860.45
age	216.36
sexmale	-109.97
bmi	287.76
children1	1106.34
children2	2086.97
children3	1845.19
children4	3780.21
children5	1869.27
smokeryes	21610.64
regionnorthwest	-354.14
regionsoutheast	-677.88
regionsouthwest	-733.51

Best lambda for Ridge: 900.663

Table 7: Ridge regression coefficients at optimal λ .

Variable	Coefficient
(Intercept)	-8804.57
age	223.93
sexmale	0
bmi	268.71
children1	309.03
children2	1227.66
children3	831.83
children4	2290.43
children5	0
smokeryes	22815.34
regionnorthwest	0
regionsoutheast	0
regionsouthwest	0

Best lambda for LASSO: 198.608

Table 8: LASSO regression coefficients at optimal λ . Zeros indicate variables that were shrunk to zero.